



Technical Service Bulletin

Technical Service Bulletin: TSB120297

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QSK19/38/50/60 CM2150 C3 Injector Misfire, Runs Rough, Cold Cylinder, or Low Exhaust Gas Temperature Fault Code Countermeasure

QSK19/38/50/60 CM2150 C3 Injector Misfire, Runs Rough, Cold Cylinder, or Low Exhaust Gas Temperature Fault Code Countermeasure

Core Issue

This document is published as a countermeasure to reduce customer downtime. A project was launched on C3 Modular Common Rail System (MCRS) injectors to identify the cause of the issue. If a C3 MCRS injector (Part Number 2867147, 2867148, or 2867149) is diagnosed as causing a misfire, rough running, cold cylinder, or low exhaust gas temperature fault code, the flow limiter in the injector may have closed, which stops fueling to that cylinder. Previous generation injectors would reset the flow limiter after the key was turned OFF and the pressure bled down, but the C3 fuel system retains fuel pressure indefinitely, which does not allow the flow limiter to reset.

Confirmation

Use the engine performance troubleshooting tree to confirm which injector is causing the issue.

Relieve fuel rail pressure to reset the flow limiters in the injectors by following the steps below.

1. Use Procedure 006-051 in the respective service manual to relieve the fuel rail pressure and then tighten the plug.
2. Use INSITE™ electronic service tool to confirm that the fuel rail pressure is less than 689.48 kPa [100 psi], which should allow the flow limiter to reset.
3. Start the engine to see if the customer complaint has been resolved.
 - If relieving the rail pressure did not resolve the issue, follow the normal steps of the engine performance troubleshooting tree.
 - If relieving the fuel rail pressure resolved the issue, monitor fuel rail pressure and shut down the engine. Reconnect to the engine control module (ECM)) and confirm rail pressure is being held (<100 bar drop in 5 minutes).
 - If pressure is held, see the Resolution section below.
 - If pressure drops, there is a leak in the system that **must** be corrected. Possible sources could be the fuel pressure relief valve or injectors.

Resolution

Update the engine's ECM calibrations to the new software as listed in the tables below.

A new feature has been added to the ECM calibration software to make the system more robust to fuel rail pressure signal noise. This new feature was included on software phase 25.2.1.04 (Industrial) ECM calibrations and will be included in future software releases. Any calibration with software prior to 25.2.1.04 (Industrial)) does **not** have this feature and is more susceptible to noise in the fuel rail pressure signal. Newer revisions to the ECM codes shown below will include the new feature.

When updating calibrations in an ECM on QSK38/50/60 engines, do **not** mix software revisions between the parent, secondary 1, or secondary 2 modules. All ECMs should use calibrations based on the same software revision.

Refer to the table below for the old and new software ECM codes that have the new feature.

QSK19 CM2150 Industrial	
New Software	Old Software
Phase 25.2.1.4	Phase 25.2.0.10
AQ40088.10	AQ40088.09
AQ40089.09	AQ40089.08
AQ40090.09	AQ40090.08
AQ40132.03	AQ40132.02
AQ40126.05	AQ40126.04
AQ40127.05	AQ40127.04
AQ40133.03	AQ40133.02
AQ40134.03	AQ40134.02
AQ40135.03	AQ40135.02
AQ40137.08	AQ40137.07
AQ40138.09	AQ40138.08
AQ40139.05	AQ40139.04
AQ40140.05	AQ40140.04
AQ40142.05	AQ40142.04
AQ40143.05	AQ40143.04
AQ40144.06	AQ40144.05
AQ40145.06	AQ40145.05
AQ40146.07	AQ40146.06
AQ40147.05	AQ40147.04
AQ40148.04	AQ40148.03
AQ40149.05	AQ40149.04
AQ40150.06	AQ40150.05
AQ40152.06	AQ40152.05
AQ40159.07	AQ40159.06
AQ40160.07	AQ40160.06

QSK19 CM2150 Industrial	
New Software	Old Software
AQ40153.05	AQ40153.04
AQ40156.05	AQ40156.04
AQ40157.05	AQ40157.04
AQ40169.07	AQ40169.06
AQ40163.06	AQ40163.05
AQ40164.06	AQ40164.05
AQ40165.06	AQ40165.05
AQ40170.04	AQ40170.03
AQ40171.04	AQ40171.03
AQ40166.06	AQ40166.05
AQ40167.06	AQ40167.05
AQ40168.04	AQ40168.03
AQ40172.05	AQ40172.04
AQ40227.04	AQ40227.03
AQ40173.06	AQ40173.05
AQ40174.06	AQ40174.05
AQ40175.06	AQ40175.05
AQ40197.06	AQ40197.05
AQ40198.06	AQ40198.05
AQ40178.06	AQ40178.05
AQ40179.06	AQ40179.05
AQ40180.07	AQ40180.06
AQ40181.06	AQ40181.05
AQ40182.06	AQ40182.05
AQ40183.05	AQ40183.04
AQ40194.05	AQ40194.04
AQ40195.04	AQ40195.03
AQ40196.05	AQ40196.04
AQ40184.04	AQ40184.03
AQ40185.03	AQ40185.02
AQ40186.04	AQ40186.03
AQ40187.06	AQ40187.05
AQ40188.04	AQ40188.03
AQ40189.04	AQ40189.03
AQ40190.03	AQ40190.02

QSK19 CM2150 Industrial	
New Software	Old Software
AQ40191.03	AQ40191.02
AQ40192.05	AQ40192.04
AQ40193.05	AQ40193.04
AQ40200.06	AQ40200.05
AQ40202.05	AQ40202.04
AQ40203.05	AQ40203.04
AQ40204.05	AQ40204.04
AQ40205.04	AQ40205.03
AQ40206.04	AQ40206.03
AQ40207.04	AQ40207.03
AQ40218.04	AQ40218.03
AQ40219.04	AQ40219.03
AQ40220.06	AQ40220.05
AQ40208.04	AQ40208.03
AQ40209.04	AQ40209.03
AQ40210.04	AQ40210.03
AQ40211.04	AQ40211.03
AQ40212.04	AQ40212.03
AQ40213.03	AQ40213.02
AQ40214.03	AQ40214.02
AQ40215.04	AQ40215.03
AQ40216.04	AQ40216.03
AQ40217.04	AQ40217.03
AQ40221.03	AQ40221.02
AQ40222.05	AQ40222.04
AQ40223.04	AQ40223.03
AQ40224.05	AQ40224.04
AQ40228.05	AQ40228.04
AQ40225.05	AQ40225.04
AQ40229.05	AQ40229.04
AQ40226.07	AQ40226.06
AQ40230.04	AQ40230.03
AQ40232.05	AQ40232.04
AQ40234.04	AQ40234.03
AQ40235.02	AQ40235.01

QSK19 CM2150 Industrial		
New Software		Old Software
AQ40233.04		AQ40233.03
AQ40239.05		AQ40239.04
AQ40241.05		AQ40241.04
AQ40240.04		AQ40240.03
AQ40245.04		AQ40245.03
AQ40248.02		AQ40248.01
AQ40249.02		AQ40249.01
AQ40250.03		AQ40250.02
AQ40251.02		AQ40251.01
AQ40252.02		AQ40252.01
AQ40247.05		AQ40247.04
AQ40253.04		AQ40253.03
AQ40254.04		AQ40254.03
AQ40267.01		AQ40267.00
AQ40269.03		AQ40269.02
AQ40270.03		AQ40270.02
AQ40262.02		AQ40262.01
AQ40263.01		AQ40263.00
AQ40264.01		AQ40264.00
AQ40258.02		AQ40258.01
AQ40259.01		AQ40259.00
AQ40260.01		AQ40260.00
AQ40261.02		AQ40261.01
AQ40277.01		AQ40277.00
AQ40278.01		AQ40278.00
AQ40279.01		AQ40279.00
QSK38 CM2150 Industrial		
Parent		
New Software	Old Software	
Phase 25.2.1.4	Phase 25.2.0.10	
AQ60215.11	AQ60215.10	
AQ60231.05	AQ60231.04	
AQ60263.04	AQ60263.03	
AQ60264.04	AQ60264.03	
AQ60265.04	AQ60265.03	

QSK38 CM2150 Industrial		
Parent		
AQ60268.04	AQ60268.03	
AQ60269.03	AQ60269.02	
AQ60282.06	AQ60282.05	
AQ60283.03	AQ60283.02	
AQ60284.05	AQ60284.04	
AQ60505.01	AQ60505.00	
AQ60300.06	AQ60300.05	
AQ60319.07	AQ60319.06	
AQ60339.05	AQ60339.04	
AQ60340.06	AQ60340.05	
AQ60351.03	AQ60351.02	
AQ60408.02	AQ60408.01	
AQ60430.02	AQ60430.01	
AQ60409.02	AQ60409.01	
AQ60393.03	AQ60393.02	
AQ60396.03	AQ60396.02	
AQ60434.02	AQ60434.01	
AQ60472.03	AQ60472.02	
AQ60435.02	AQ60435.01	
AQ60436.02	AQ60436.01	
AQ60437.02	AQ60437.01	
AQ60474.01	AQ60474.00	
AQ60459.06	AQ60459.05	
AQ60478.02	AQ60478.01	
AQ60522.01	AQ60522.00	
AQ60523.01	AQ60523.00	
AQ60524.01	AQ60524.00	
AQ60535.01	AQ60535.00	
QSK38 CM2150 Industrial		
Secondary 1		
New Software	Old Software	
Phase 25.2.1.4	Phase 25.2.0.10	
AR60190.06	AR60190.05	
AR60205.05	AR60205.04	
AR60227.04	AR60227.03	

QSK38 CM2150 Industrial		
Secondary 1		
AR60228.04	AR60228.03	
AR60229.04	AR60229.03	
AR60230.03	AR60230.02	
AR60231.03	AR60231.02	
AR60250.03	AR60250.02	
AR60251.03	AR60251.02	
AR60252.03	AR60252.02	
AR60334.02	AR60334.01	
AR60335.02	AR60335.01	
AR60336.02	AR60336.01	
AR60327.03	AR60327.02	
AR60374.03	AR60374.02	
AR60380.01	AR60380.00	
AR60389.01	AR60389.00	
AR60390.01	AR60390.00	
AR60391.01	AR60391.00	
AR60403.01	AR60403.00	

QSK50/60 CM2150 Industrial		
Parent		
New Software	Old Software	
Phase 25.2.1.4	Phase 25.2.0.10	
AQ60218.12	AQ60218.11	
AQ60217.10	AQ60217.09	
AQ60220.13	AQ60220.12	
AQ60221.11	AQ60221.10	
AQ60232.05	AQ60232.04	
AQ60236.09	AQ60236.08	
AQ60237.11	AQ60237.10	
AQ60271.04	AQ60271.03	
AQ60272.04	AQ60272.03	
AQ60273.04	AQ60273.03	
AQ60274.05	AQ60274.04	
AQ60275.05	AQ60275.04	
AQ60276.05	AQ60276.04	

QSK50/60 CM2150 Industrial

Parent

AQ60278.05	AQ60278.04
AQ60279.05	AQ60279.04
AQ60238.11	AQ60238.10
AQ60317.08	AQ60317.07
AQ60287.07	AQ60287.06
AQ60288.07	AQ60288.06
AQ60337.07	AQ60337.06
AQ60302.08	AQ60302.07
AQ60305.10	AQ60305.09
AQ60421.05	AQ60421.04
AQ60289.10	AQ60289.09
AQ60303.11	AQ60303.10
AQ60252.07	AQ60252.06
AQ60253.11	AQ60253.10
AQ60255.04	AQ60255.03
AQ60285.04	AQ60285.03
AQ60254.14	AQ60254.13
AQ60295.06	AQ60295.05
AQ60296.06	AQ60296.05
AQ60292.05	AQ60292.04
AQ60293.05	AQ60293.04
AQ60294.04	AQ60294.03
AQ60297.03	AQ60297.02
AQ60309.06	AQ60309.05
AQ60310.06	AQ60310.05
AQ60301.07	AQ60301.06
AQ60304.08	AQ60304.07
AQ60318.06	AQ60318.05
AQ60314.08	AQ60314.07
AQ60312.07	AQ60312.06
AQ60313.06	AQ60313.05
AQ60315.06	AQ60315.05
AQ60316.08	AQ60316.07
AQ60359.06	AQ60359.05
AQ60360.05	AQ60360.04

QSK50/60 CM2150 Industrial

Parent

AQ60361.05	AQ60361.04
AQ60362.05	AQ60362.04
AQ60320.05	AQ60320.04
AQ60326.08	AQ60326.07
AQ60327.11	AQ60327.10
AQ60328.08	AQ60328.07
AQ60329.09	AQ60329.08
AQ60330.08	AQ60330.07
AQ60451.02	AQ60451.01
AQ60429.05	AQ60429.04
AQ60325.07	AQ60325.06
AQ60401.08	AQ60401.07
AQ60331.05	AQ60331.04
AQ60341.08	AQ60341.07
AQ60342.07	AQ60342.06
AQ60335.08	AQ60335.07
AQ60343.06	AQ60343.05
AQ60344.04	AQ60344.03
AQ60345.04	AQ60345.03
AQ60338.08	AQ60338.07
AQ60363.06	AQ60363.05
AQ60364.05	AQ60364.04
AQ60365.05	AQ60365.04
AQ60366.05	AQ60366.04
AQ60378.06	AQ60378.05
AQ60379.05	AQ60379.04
AQ60380.05	AQ60380.04
AQ60386.06	AQ60386.05
AQ60381.05	AQ60381.04
AQ60352.07	AQ60352.06
AQ60373.06	AQ60373.05
AQ60374.05	AQ60374.04
AQ60375.05	AQ60375.04
AQ60376.05	AQ60376.04
AQ60377.05	AQ60377.04

QSK50/60 CM2150 Industrial

Parent

AQ60367.05	AQ60367.04
AQ60368.05	AQ60368.04
AQ60369.05	AQ60369.04
AQ60370.05	AQ60370.04
AQ60371.05	AQ60371.04
AQ60387.06	AQ60387.05
AQ60388.05	AQ60388.04
AQ60389.05	AQ60389.04
AQ60390.05	AQ60390.04
AQ60395.07	AQ60395.06
AQ60391.07	AQ60391.06
AQ60392.05	AQ60392.04
AQ60402.06	AQ60402.05
AQ60398.08	AQ60398.07
AQ60399.06	AQ60399.05
AQ60404.06	AQ60404.05
AQ60397.05	AQ60397.04
AQ60410.05	AQ60410.04
AQ60412.02	AQ60412.01
AQ60413.03	AQ60413.02
AQ60441.04	AQ60441.03
AQ60416.03	AQ60416.02
AQ60442.03	AQ60442.02
AQ60414.03	AQ60414.02
AQ60415.03	AQ60415.02
AQ60433.03	AQ60433.02
AQ60417.05	AQ60417.04
AQ60431.05	AQ60431.04
AQ60420.02	AQ60420.01
AQ60443.05	AQ60443.04
AQ60440.03	AQ60440.02
AQ60444.04	AQ60444.03
AQ60465.02	AQ60465.01
AQ60466.02	AQ60466.01
AQ60467.02	AQ60467.01

QSK50/60 CM2150 Industrial

Parent

AQ60517.01	AQ60517.00
AQ60463.03	AQ60463.02
AQ60464.03	AQ60464.02
AQ60446.04	AQ60446.03
AQ60445.03	AQ60445.02
AQ60453.03	AQ60453.02
AQ60454.03	AQ60454.02
AQ60455.03	AQ60455.02
AQ60458.03	AQ60458.02
AQ60456.04	AQ60456.03
AQ60475.03	AQ60475.02
AQ60476.03	AQ60476.02
AQ60477.02	AQ60477.01
AQ60469.02	AQ60469.01
AQ60479.01	AQ60479.00
AQ60480.01	AQ60480.00
AQ60481.02	AQ60481.01
AQ60490.04	AQ60490.03
AQ60491.01	AQ60491.00
AQ60492.01	AQ60492.00
AQ60493.02	AQ60493.01
AQ60482.02	AQ60482.01
AQ60489.01	AQ60489.00
AQ60484.01	AQ60484.00
AQ60509.02	AQ60509.01
AQ60510.01	AQ60510.00
AQ60511.01	AQ60511.00
AQ60512.01	AQ60512.00
AQ60506.01	AQ60506.00
AQ60507.01	AQ60507.00
AQ60508.01	AQ60508.00
AQ60501.01	AQ60501.00
AQ60502.01	AQ60502.00
AQ60503.01	AQ60503.00
AQ60499.02	AQ60499.01

QSK50/60 CM2150 Industrial		
Parent		
AQ60498.02	AQ60498.01	
AQ60500.02	AQ60500.01	
AQ60504.01	AQ60504.00	
AQ60526.01	AQ60526.00	
AQ60527.01	AQ60527.00	
AQ60528.01	AQ60528.00	
AQ60530.01	AQ60530.00	
AQ60531.01	AQ60531.00	
AQ60534.01	AQ60534.00	
QSK50/60 CM2150 Industrial		
Secondary 1		
New Software	Old Software	
Phase 25.2.1.4	Phase 25.2.0.10	
AR60195.04	AR60195.03	
AR60193.05	AR60193.04	
AR60199.05	AR60199.04	
AR60201.07	AR60201.06	
AR60206.05	AR60206.04	
AR60210.04	AR60210.03	
AR60232.04	AR60232.03	
AR60234.04	AR60234.03	
AR60236.04	AR60236.03	
AR60238.05	AR60238.04	
AR60240.05	AR60240.04	
AR60242.05	AR60242.04	
AR60246.05	AR60246.04	
AR60248.05	AR60248.04	
AR60212.03	AR60212.02	
AR60287.02	AR60287.01	
AR60255.03	AR60255.02	
AR60257.03	AR60257.02	
AR60307.04	AR60307.03	
AR60275.04	AR60275.03	
AR60279.04	AR60279.03	

QSK50/60 CM2150 Industrial

Secondary 1

AR60349.02	AR60349.01
AR60259.03	AR60259.02
AR60277.04	AR60277.03
AR60221.02	AR60221.01
AR60253.02	AR60253.01
AR60219.06	AR60219.05
AR60265.03	AR60265.02
AR60267.03	AR60267.02
AR60269.03	AR60269.02
AR60271.02	AR60271.01
AR60283.02	AR60283.01
AR60285.04	AR60285.03
AR60321.03	AR60321.02
AR60323.03	AR60323.02
AR60325.03	AR60325.02
AR60293.04	AR60293.03
AR60295.04	AR60295.03
AR60297.02	AR60297.01
AR60299.04	AR60299.03
AR60301.02	AR60301.01
AR60358.02	AR60358.01
AR60328.05	AR60328.04
AR60303.04	AR60303.03
AR60309.04	AR60309.03
AR60311.04	AR60311.03
AR60313.02	AR60313.01
AR60315.02	AR60315.01
AR60317.02	AR60317.01
AR60330.02	AR60330.01
AR60337.02	AR60337.01
AR60339.02	AR60339.01
AR60341.03	AR60341.02
AR60343.02	AR60343.01
AR60347.02	AR60347.01
AR60354.02	AR60354.01

QSK50/60 CM2150 Industrial		
Secondary 1		
AR60364.02	AR60364.01	
AR60366.02	AR60366.01	
AR60368.02	AR60368.01	
AR60360.02	AR60360.01	
AR60362.02	AR60362.01	
AR60372.02	AR60372.01	
AR60392.01	AR60392.00	
AR60383.01	AR60383.00	
AR60405.01	AR60405.00	
QSK50/60 CM2150 Industrial		
Secondary 2		
New Software	Old Software	
Phase 25.2.1.4	Phase 25.2.0.10	
AR60196.04	AR60196.03	
AR60194.05	AR60194.04	
AR60200.05	AR60200.04	
AR60202.07	AR60202.06	
AR60207.05	AR60207.04	
AR60233.04	AR60233.03	
AR60235.04	AR60235.03	
AR60237.04	AR60237.03	
AR60239.05	AR60239.04	
AR60241.05	AR60241.04	
AR60243.06	AR60243.05	
AR60247.05	AR60247.04	
AR60249.05	AR60249.04	
AR60213.03	AR60213.02	
AR60211.04	AR60211.03	
AR60288.02	AR60288.01	
AR60256.03	AR60256.02	
AR60258.03	AR60258.02	
AR60308.04	AR60308.03	
AR60276.04	AR60276.03	
AR60280.04	AR60280.03	

QSK50/60 CM2150 Industrial

Secondary 2

AR60350.02	AR60350.01
AR60260.03	AR60260.02
AR60278.04	AR60278.03
AR60222.02	AR60222.01
AR60254.02	AR60254.01
AR60220.06	AR60220.05
AR60266.03	AR60266.02
AR60268.03	AR60268.02
AR60270.03	AR60270.02
AR60272.02	AR60272.01
AR60284.02	AR60284.01
AR60286.04	AR60286.03
AR60322.03	AR60322.02
AR60324.03	AR60324.02
AR60326.03	AR60326.02
AR60294.04	AR60294.03
AR60296.04	AR60296.03
AR60298.02	AR60298.01
AR60300.04	AR60300.03
AR60302.02	AR60302.01
AR60359.02	AR60359.01
AR60355.02	AR60355.01
AR60329.05	AR60329.04
AR60304.04	AR60304.03
AR60310.04	AR60310.03
AR60312.04	AR60312.03
AR60314.02	AR60314.01
AR60316.02	AR60316.01
AR60318.02	AR60318.01
AR60331.02	AR60331.01
AR60338.02	AR60338.01
AR60340.02	AR60340.01
AR60342.03	AR60342.02
AR60344.02	AR60344.01
AR60348.02	AR60348.01

QSK50/60 CM2150 Industrial		
Secondary 2		
AR60365.02	AR60365.01	
AR60367.02	AR60367.01	
AR60369.02	AR60369.01	
AR60361.02	AR60361.01	
AR60363.02	AR60363.01	
AR60373.02	AR60373.01	
AR60393.01	AR60393.00	
AR60394.01	AR60394.00	
AR60395.01	AR60395.00	
AR60384.01	AR60384.00	
AR60406.01	AR60406.00	

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Document History

Date	Details
2012-11-26	Module Created
2013-3-25	Revise Resolution Section
2013-4-24	Update
2013-8-19	Updating to include Step 6 countermeasure
2013-9-9	Remove Secondary 1